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1  (*
2      CS51 Lab 7
3      Modules & Functors
4
5      A module for colors and color names
6
7  The representation for colors in this module is really obscure (and
8  arguably unnecessarily so). By the way, it also has some bugs so it
9  doesn't pass all the unit tests. No need to debug it though, or even
10 read it. You'll be replacing it wholesale with your own hopefully
11 simpler implementation, after constructing an appropriate signature
12 for the module in the file `color.mli`. *)
13
14  (*
15      SOLUTION
16
17  In this implementation, we simply use int triples for the RGB
18  channels. By keeping the data in a more structured form, all of the
19  conversions are simplified.
20
21  You might have used a record type instead of a tuple, for instance,
22
23      type color = { red : int; green : int; blue : int } ;;
24
25  That could work too. And the user of the module (the file
26  `lab7_part2.ml`) *wouldn't be able to tell the difference*, just like
27  it can't tell the difference with the horribly obfuscated
28  implementation in the original version of `color.ml`.
29
30  You'll notice that we've moved all of the documentation on module
31  elements (types, functions, and the like) into the `.mli` file, since
32  it's the signature file that codifies what users of the module "need
33  to know". *)
34
35  type color = int * int * int ;;
36
37  type color_name =
38      | Red
39      | Green
40      | Blue
41      | Orange
42      | Yellow
43      | Indigo
44      | Violet ;;
45
46  let to_color (r : int) (g : int) (b : int) : color =
47      r, g, b ;;
48
49  let red_channel  ( r, _g, _b : color) : int = r ;;
50  let green_channel (_r,  g, _b : color) : int = g ;;
51  let blue_channel  (_r, _g,  b : color) : int = b ;;
52

```

```

53 let color_named (name : color_name) : color =
54   match name with
55   | Red    -> to_color 255  0  0
56   | Green  -> to_color  0 255  0
57   | Blue   -> to_color  0  0 255
58   | Orange -> to_color 255 165  0
59   | Yellow -> to_color 255 255  0
60   | Indigo -> to_color  75  0 130
61   | Violet -> to_color 240 130 240

```